

Intent-to-API Orchestration for Future Telco Networks: A CAPIF-Aware, AI-Native Platform with Empirical Reliability Evaluation

Türkan Doğa Durak

Centre Tecnològic de Telecomunicacions de Catalunya (CTTC)

Castelldefels, Spain

turkandoga.durak@cttc.es

Abstract—Future Internet architectures demand programmable network exposure frameworks that bridge high-level operator intents to concrete telecom API invocations without manual payload composition. This paper presents the design, implementation, and empirical evaluation of a cloud-native, CAPIF-aware telco orchestration platform combining 3GPP CAPIF governance, CAMARA-aligned service semantics, NEF-backed telecom realisation, and LLM-assisted planning under a three-layer deterministic validator. A two-phase schema-grounded planning pipeline separates service routing from payload generation, eliminating payload-level LLM errors entirely. Evaluation on a bilingual 60-entry ground-truth dataset using six formally defined reliability metrics yields IWSR=100 % (deterministic) and IWSR=95 % (LLM-assisted, Claude Haiku 4.5), with perfect Schema Compliance (SCR=100 %) and Semantic Validity (SVR=100 %) in both modes and a Stability Score of SS=0.892. A preliminary multi-provider comparison shows that the pluggable multi-LLM dispatcher (Gemini 2.5 Flash, GPT-4o-mini, Claude Haiku 4.5) delivers consistent SCR=SVR=100 % across all providers, confirming that the schema-grounding mechanism is provider-agnostic. All LLM failures are Phase 1 service-routing errors on deliberately ambiguous intents; no payload schema violation occurs in either mode, validating the effectiveness of schema-grounded Phase 2 prompting. The platform and evaluation dataset are publicly released as open-source artefacts providing a reproducible baseline for intent-based orchestration research.

Index Terms—intent-based networking, CAPIF, CAMARA, NEF, 5G/B5G service exposure, LLM-assisted orchestration, reliability evaluation, zero-touch automation, cloud-native, future internet

I. INTRODUCTION

The emergence of Beyond-5G (B5G) and 6G network architectures elevates programmable service exposure from an optional feature to a foundational design principle of the Future Internet. Standardised frameworks such as 3GPP CAPIF (TS 23.222) [1] and the CAMARA project [2] define the governance, discovery, and semantic layers enabling vertical industries to consume network capabilities through well-defined APIs. NEF-based provider stacks [4] bridge those interfaces to the underlying data plane, resolving identifiers, enforcing policies, and returning structured monitoring results.

In this layered model, orchestration logic must translate natural-language intents into precisely structured, multi-field API payloads that comply with strict telecom domain constraints: E.164 phone formats, enumerated QoS profiles, duration bounds, and live CAPIF registry membership. Manual payload composition at this level is error-prone and scales poorly, motivating automated intent-to-API translation. Large language models (LLMs) offer a compelling approach [6], [8]; however, telecom API payloads impose hard correctness constraints that probabilistic generation cannot guarantee without external enforcement [7], [9]. Hallucinated service identifiers, malformed phone numbers, or invalid QoS profile names would silently propagate to production networks if not intercepted.

This paper addresses the gap between LLM planning capability and telecom-grade reliability. The main contributions are:

- a working, open-source CAPIF-aware telco orchestration platform operating against live CAMARA providers in a multi-container Docker deployment;
- a two-phase schema-grounded LLM planner with a pluggable multi-provider dispatcher (Gemini, OpenAI, Anthropic) and live CAPIF discovery injection;
- a three-layer deterministic validator (L1 Pydantic schema, L2 telecom semantic policy, L3 CAPIF registry) with a bounded feedback correction loop that guarantees payload correctness independent of LLM quality;
- a bilingual 60-entry ground-truth evaluation dataset with difficulty stratification and multi-country phone coverage;
- six formally defined reliability metrics (IWSR, SCR, SVR, HR, SS, RRF) constituting a reusable evaluation framework for intent-to-API orchestration systems.

The remainder of the paper is organised as follows. Section II surveys related work. Section III describes the platform architecture and deployment. Section IV details the LLM planning and validation layer. Section V presents the evaluation framework and results. Section VI discusses findings and limitations. Section VII concludes.

II. RELATED WORK

A. Intent-Based Networking and Orchestration

Intent-based networking (IBN) has been explored extensively at the NFV orchestration level. Rafiq et al. [10] demonstrated deterministic intent-to-descriptor translation across multiple orchestrators; Saraiva de Sousa et al. [12] identified intent abstraction as a key direction for adaptive service management; and Coronado et al. [11] positioned intent-driven closed-loop control as foundational to zero-touch 6G management. These works operate at the orchestration lifecycle level and generally assume structured intent inputs, not natural language.

B. 5G Network Exposure Infrastructure

Molina Sanchez et al. [3] implemented CAPIF Core Function as an open-source microservice (OpenCAPIF), enabling community experimentation with 3GPP-compliant API exposure. Fragkos et al. [4] introduced NEFSim as an open NEF experimentation platform covering the full 3GPP service exposure stack. Turkovic et al. [5] demonstrated 3GPP-compliant CAMARA QoS and Location API integration in a real CAM testbed, measuring end-to-end latency of 230–340 ms. Our work builds on these open platforms as provider components and focuses on the intent translation and orchestration layer above them.

C. LLMs for Telecom and Structured Output

Maatouk et al. [6] identified hallucination and output inconsistency as primary risks for LLM deployment in telecom operations. Liu et al. [9] benchmarked structured-output compliance of leading LLMs and found that GPT-4 achieves an average compliance score of only 0.40 on JSON schema tasks. Lee et al. [7] confirmed that multi-turn planning tasks significantly degrade LLM pass rates in telecom agent scenarios. Zeng et al. [8] demonstrated near-perfect workflow generation when LLMs are grounded in structured API descriptions, the approach adopted in Phase 2 of our planner.

Research Gap. To the best of our knowledge, no prior work provides end-to-end empirical reliability evaluation of LLM-based natural-language intent-to-API translation within a live CAPIF-compliant 5G service exposure environment, operating under simultaneous schema, semantic, and registry constraints, and reporting a standardised six-metric reliability framework on a bilingual ground-truth dataset.

III. PLATFORM ARCHITECTURE

A. System Overview and Components

The platform comprises nine functional components organised into four logical tiers: a user-facing *interface tier*, a *control-plane tier* containing orchestration logic, a *provider tier* hosting CAMARA API realisations, and a *persistence tier* backed by MongoDB. Fig. 1 depicts the complete cloud-native deployment topology; Table I summarises each component’s responsibility.

TABLE I
IMPLEMENTED PLATFORM COMPONENTS

Component	Responsibility
Operator Interface	Product-style landing page with embedded OpenAPI console
FastAPI Backend	Route handling, orchestration, validation, execution
Intent Engine	Deterministic keyword-based routing (baseline mode)
LLM Planner	Two-phase schema-grounded structured planning
Multi-Layer Validator	L1 Pydantic schema, L2 semantic policy, L3 CAPIF registry
Feedback Engine	Bounded correction loop (max 3 iterations)
CAPIF Client	Invoker onboarding, mTLS discovery, provider publication
Provider Layer	QoS, Location Retrieval, QoS Provisioning, QoS Profiles
Persistence Layer	MongoDB-backed NEF data with automatic initialisation

TABLE II
MAIN BACKEND ENDPOINTS

Method	Path	Purpose
GET	/health	Runtime health and mode summary
GET	/catalog/services	CAMARA-aligned capability catalog
POST	/orchestrate	Intent-to-execution planning
GET	/capif/status	CAPIF connectivity check
POST	/capif/invokers/onboard	Live invoker onboarding
GET	/capif/discover	Live CAPIF service discovery
POST	/capif/publish/catalog	Publish all catalog services

B. Cloud-Native Multi-Container Deployment

The platform is packaged as a Docker Compose stack comprising five networked services: **backend** (FastAPI, port 8000), **OpenCAPIF Core Function** (3GPP-compliant register and discovery), **NEF provider** (port 8080), **camara-QoS provider** (port 8090), and **NEF-QoS provider** (port 8585). All containers share a Docker bridge network; service discovery across providers is handled at runtime by the CAPIF client querying the Core Function’s discovery endpoint rather than through static configuration. A named Docker volume persists MongoDB data across restarts, and an initialisation script seeds location, IMSI, and polygon records on first deployment. The platform includes 65 automated unit tests across three pytest modules, enabling CI/CD integration without any LLM API key or live CAPIF connection as external dependencies.

C. CAPIF Integration and CAMARA Service Realisation

The backend acts as a CAPIF **Invoker**, following the 3GPP CAPIF-2e reference point: (1) authenticate against the register

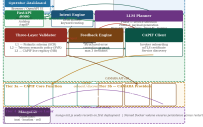


Fig. 1. Cloud-native system architecture of the Mini Telco Orchestration Platform. Tier 1 (Operator Dashboard) submits intents via REST to Tier 2 (Control Plane), where the FastAPI backend delegates to the Intent Engine (deterministic) or the two-phase LLM Planner. All payloads traverse a three-layer validator (L1–L3) with bounded feedback correction. Tier 3a (OpenCAPIF Core Function at ports :8084/:8080 with Vault :8200 and nginx :443) provides mTLS onboarding and live service discovery. Tier 3b (CAMARA Providers at :8002, :8003, :8585) hosts QoS, Location Retrieval, and NEF-QoS. Tier 4 (MongoDB) persists IMSI, location, and cell-polygon records; `mongo-init.js` seeds records on first deployment. All containers share the `capif-network` Docker bridge.

service; (2) generate an RSA private key and CSR locally; (3) submit the onboarding request to the CAPIF Core Function; (4) receive an `apiInvokerId` and certificate chain; (5) use mTLS for all subsequent discovery and execution. CAMARA providers act as **API Exposing Functions (AEFs)**, publishing OpenAPI metadata through the CAPIF-4 reference point. This three-role governance model ensures that every API invocation is CAPIF-mediated: services are visible to the orchestration layer only after publication, enabling the L3 validator to detect hallucinated service identifiers via live registry queries.

The platform exposes four CAMARA-standardised orchestration targets: *QoS* for temporary QoS session creation (CAMARA QoS v1.1.0); *QoS Profiles* for available profile enumeration; *QoS Provisioning* for long-lived QoS assignment via 3GPP `AsSessionWithQoS` (TS 29.122); and *Location Retrieval* for device location access through NEF-backed `MonitoringEvent` subscriptions (CAMARA Location-Retrieval v0.5). The FRONT-research-group provider stacks [5] are included as Git submodules, enabling version-tracked integration.

IV. LLM-ASSISTED ORCHESTRATION

A. Two-Phase Schema-Grounded Planning

The LLM planner implements a two-phase pipeline that separates service routing from payload generation, limiting each call to a well-scoped subtask with a clearly defined schema target (Fig. 2).

Phase 1 — Service Selection: The LLM receives (a) the user intent in natural language, (b) a schema-grounded summary of all CAMARA services extracted by parsing YAML OpenAPI

descriptors at startup, and (c) optionally, live CAPIF discovery results. It returns a JSON object specifying `service_id`, `operation_id`, `method`, `path`, and a `rationale` field for downstream traceability.

Phase 2 — Payload Generation: Given the Phase 1 service selection, the LLM receives the full CAMARA OpenAPI schema for that specific operation extracted at runtime. It returns a complete, schema-aligned JSON payload matching the operation’s request body specification. Grounding the payload prompt in the actual operation schema — rather than a generic description — is the key mechanism achieving SCR = 100 % in LLM-assisted mode, confirming the FlowMind hypothesis [8] in a live telecom environment.

Three LLM providers are supported via a unified dispatcher: Gemini (`GEMINI_API_KEY`), OpenAI (`OPENAI_API_KEY`), and Anthropic (`ANTHROPIC_API_KEY`). The evaluation in this paper used **Claude Haiku 4.5** (temperature 0.1) at a total API cost of approximately \$0.35 across 300 two-phase call pairs (60 entries $\times$ 5 stability runs).

B. Three-Layer Deterministic Validator

The planner proposes structured actions but never executes them directly. All generated payloads pass three independent validation tiers sequentially before any provider call is made:

- **L1 — Schema (SCR):** Pydantic v2 model validation enforcing required fields, correct types, nested object structure, and field constraints (e.g., duration as positive integer). Failures produce structured error objects returned to the feedback engine.

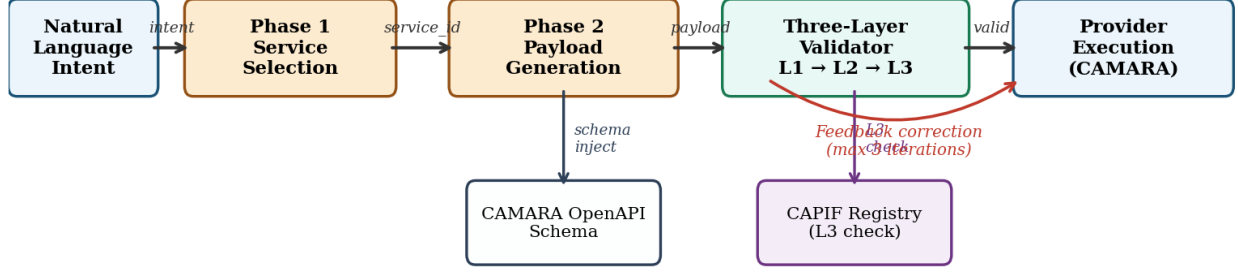


Fig. 2. Two-phase LLM planning pipeline with schema grounding and three-layer validation. Phase 1 performs service selection grounded in CAMARA OpenAPI schema summaries and optional live CAPIF discovery results. Phase 2 generates the full API payload for the selected operation, injecting the complete operation schema. All output passes through the three-layer validator (L1–L3) before execution; invalid payloads re-enter the feedback correction loop (max 3 iterations).

- **L2 — Semantic Policy (SVR):** Eight domain-specific deterministic rules: E.164 phone number format, valid QoS profile identifiers from the CAMARA enumeration, duration within 1–7200 s, IPv4 unicast address validation, operation/method consistency, and device field completeness.
- **L3 — CAPIF Registry (HR):** Live CAPIF discovery query verifying the selected `service_id` exists in the published registry. This layer fails open on CAPIF unavailability but records the check result for audit traceability.

This architectural separation between probabilistic LLM reasoning and deterministic telecom constraint enforcement is the central property enabling quantified reliability guarantees.

C. Feedback Correction Loop

When Phase 2 output fails the three-layer validator, the feedback engine constructs a structured correction prompt containing: the original intent, the rejected payload JSON, and layer-tagged error messages (e.g., [L1] `device.phoneNumber` must match E.164). The loop repeats up to `MAX_ITERATIONS=3`, after which the most recent attempt is returned with the validation error attached. Recovery metadata is recorded in a `FeedbackMetadata` object and included in the orchestration response, enabling offline analysis of correction convergence rates.

The three planning modes supported by `POST /orchestrate` are: *deterministic* (keyword routing, no LLM call), *LLM-assisted* (two-phase planner, fails explicitly on error), and *auto* (LLM primary with deterministic fallback). This design keeps the AI layer optional while preserving full functionality regardless of LLM availability.

V. RELIABILITY EVALUATION

A. Formal Metric Definitions

Intent-to-API translation is formalised as a mapping $f_\theta : \mathcal{I} \rightarrow \mathcal{W}$, where $\mathcal{I}$ is the set of natural language intents and

$\mathcal{W}$ is the set of CAPIF-compliant API workflows. Six metrics are defined:

$$\text{IWSR} = \frac{|\{w \in \mathcal{W} \mid w = w^*\}|}{|\mathcal{I}|}, \quad \text{SCR} = \frac{|\{w \models \text{Schema}\}|}{|\mathcal{W}|} \quad (1)$$

$$\text{SVR} = \frac{|\{w \models \text{Policy} \mid w \models \text{Schema}\}|}{|\{w \models \text{Schema}\}|}, \quad \text{HR} = \frac{|\{w \not\subseteq \mathcal{A}_{\text{reg}}\}|}{|\mathcal{W}|} \quad (2)$$

$$\text{SS} = 1 - \frac{d-1}{k}, \quad \text{RRF} = \frac{|\{w_c = w^* \mid w_i \not\models \text{Schema}\}|}{|\{w_i \not\models \text{Schema}\}|} \quad (3)$$

where w^* is the ground-truth workflow, $\mathcal{A}_{\text{reg}}$ is the live CAPIF registry, k is the number of repeated runs, and d is the number of distinct output tuples per entry across k runs. IWSR is the primary end-to-end correctness metric; HR captures service-selection hallucinations; SS measures output stability under repeated identical inputs; RRF is undefined when $\text{SCR} = 100\%$ (no schema failures occur on first attempt).

B. Evaluation Dataset

The ground-truth dataset (`datasets/ground_truth.json`) contains 60 manually crafted entries spanning all four supported services: 20 for QoS Session creation, 15 for Location Retrieval, 10 for QoS Profiles, 10 for QoS Provisioning, and 5 deliberate edge cases (ambiguous or boundary-condition intents). Difficulty is stratified as 24 easy, 20 medium, and 16 hard entries. Language coverage is 32 English and 28 Turkish entries, providing bilingual robustness evaluation without additional prompt engineering. Phone numbers span four country codes (+30 GR, +90 TR, +34 ES, +1 US) sourced from FRONT-research-group test fixtures, testing E.164 format validation across international prefixes. The dataset and evaluation scripts are released alongside the platform.

TABLE III
RELIABILITY METRICS: DETERMINISTIC VS. LLM-ASSISTED MODE (60 ENTRIES)

Metric	Deterministic	LLM (Claude Haiku 4.5)
IWSR	100.0 %	95.0 %
SCR	100.0 %	100.0 %
SVR	100.0 %	100.0 %
HR	0.0 %	5.0 %
SS	1.000	0.892
RRF	N/A	N/A ^a
Avg. latency	<1 ms	≈3 s

^aRRF is undefined when SCR = 100 % (the feedback loop was never triggered).

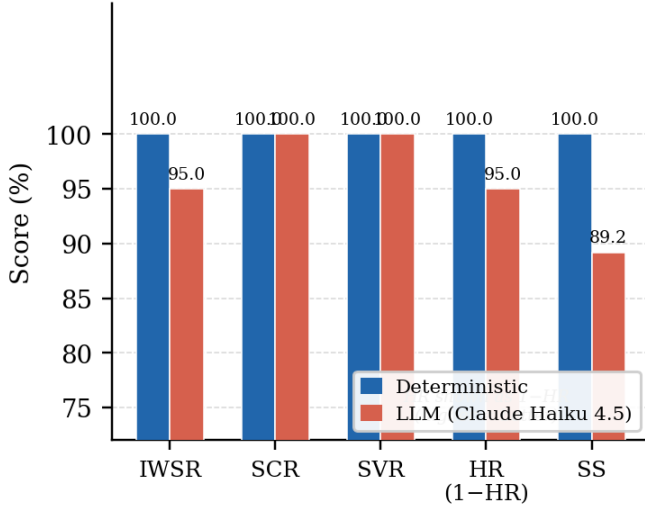


Fig. 3. Reliability metric comparison between deterministic and LLM-assisted modes across 60 evaluation entries. HR is displayed as 100 % – HR (inverted; higher is better). SCR and SVR are identical in both modes; the 5-point IWSR gap is entirely attributable to Phase 1 service-routing errors.

C. Results

Table III and Fig. 3 compare both modes across all six metrics. The deterministic mode achieves perfect scores in all metrics. The LLM-assisted mode achieves IWSR = 95 %, SCR = SVR = 100 %, HR = 5 %, and SS = 0.892. Notably, SCR and SVR are identical in both modes, confirming that Phase 2 schema grounding fully prevents payload-level LLM errors.

D. IWSR Breakdown by Dimension

Fig. 4 and Table IV detail the IWSR breakdown across three orthogonal dimensions in LLM-assisted mode.

Language: Turkish entries achieve 96.4 % vs. 93.8 % for English, a 2.6-point advantage indicating robust cross-lingual generalisation without dedicated multilingual prompt engineering. The small gap is consistent with the Turkish entries being slightly shorter and more unambiguous in phrasing.

Difficulty: Easy and medium entries achieve perfect IWSR = 100 %, while hard entries drop to 81.2 %. This confirms that difficulty stratification correctly isolates Phase 1

TABLE IV
IWSR BREAKDOWN BY DIMENSION (LLM-ASSISTED MODE)

Dim.	Category	N	IWSR
Language	English	32	93.8 %
	Turkish	28	96.4 %
Difficulty	Easy	24	100.0 %
	Medium	20	100.0 %
	Hard	16	81.2 %
Service	QoD Session	20	90.0 %
	Location Retrieval	15	100.0 %
	QoS Profiles	10	100.0 %
	QoS Provisioning	10	100.0 %
	Edge Cases	5	80.0 %

routing ambiguity: easy and medium intents use unambiguous service-specific phrasing, while hard intents deliberately exploit semantic overlap between services.

Service: Three services achieve IWSR = 100 %. QoD (90.0 %) and edge cases (80.0 %) underperform due to Phase 1 routing ambiguity between semantically similar services, as detailed in the error analysis below.

E. Error Analysis

The 5 % IWSR gap in LLM mode corresponds to 3 entries (Fig. 5), all classified as service-selection errors (HR = 5 %). Critically, in all failure cases, the payload generated for the wrongly-selected service was structurally and semantically valid (SCR = SVR = 100 %), confirming that Phase 2 schema grounding fully prevents payload-level errors.

Two failure patterns were identified. *Pattern 1 — QoD vs. QoS Provisioning ambiguity (2 entries):* Intents containing “critical communications QoS” or “emergency QoS” caused the model to select qos-provisioning (long-lived permanent assignment) instead of qod (temporary session). The model correctly identifies the QoS priority requirement but misinterprets the temporal scope from the phrasing. *Pattern 2 — Edge case boundary (1 entry):* An intent specifying a very short duration (“QoS for 30 seconds”) caused confusion between a QoD temporary session and a provisioning operation. Explicit temporal quantifiers in the Phase 1 prompt would address both patterns.

F. Multi-LLM Provider Comparison

A key design principle of the platform is that the LLM planner is provider-agnostic: a pluggable dispatcher (implemented in `backend/llm_planner.py`) selects between Gemini, OpenAI, and Anthropic at runtime via `LLM_PROVIDER` environment variable without any change to the orchestration logic. Fig. 6 and Table V compare the four operating configurations across the primary reliability metrics.

A critical observation is that SCR and SVR reach 100 % in all three LLM providers. This confirms that the schema-grounding mechanism is entirely provider-agnostic: the three-layer validator intercepts any payload error before execution regardless of which language model generated the content.

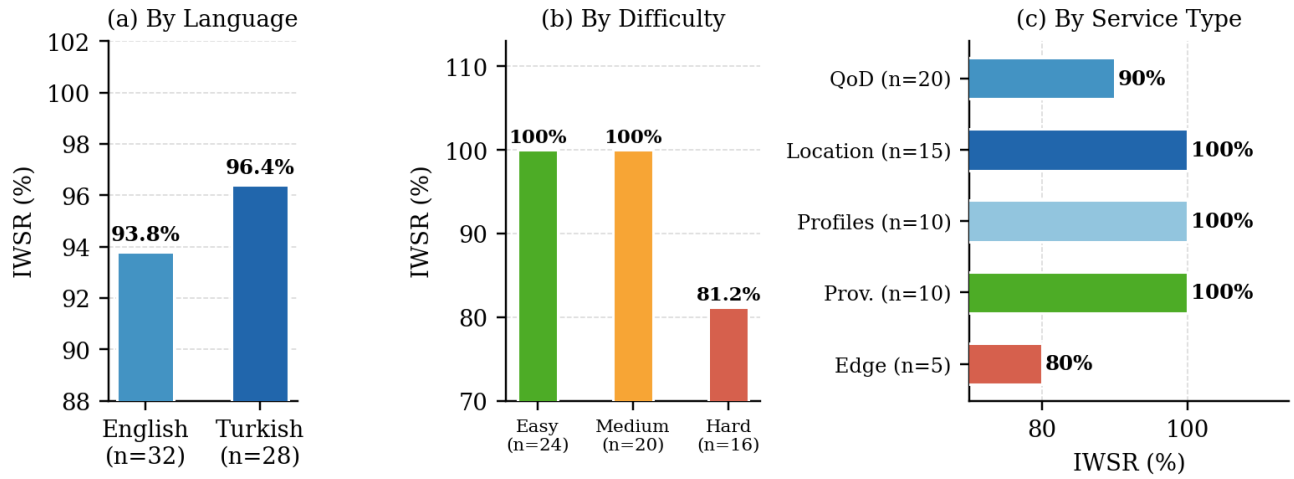


Fig. 4. IWSR breakdown by (a) language, (b) difficulty level, and (c) service type in LLM-assisted mode ($n = 60$). Turkish entries marginally outperform English (96.4 % vs. 93.8 %), demonstrating cross-lingual generalisation without additional prompt engineering. Easy and medium entries achieve perfect IWSR; hard entries and edge cases expose Phase 1 routing ambiguity.

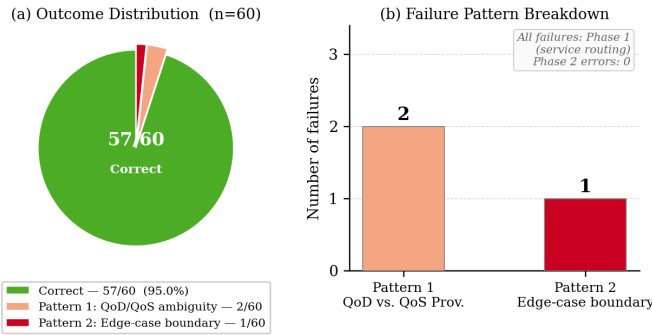


Fig. 5. LLM error analysis: (a) outcome distribution across 60 entries; (b) failure pattern breakdown. All 3 failures are Phase 1 service-routing errors; zero Phase 2 payload errors occur in any mode.

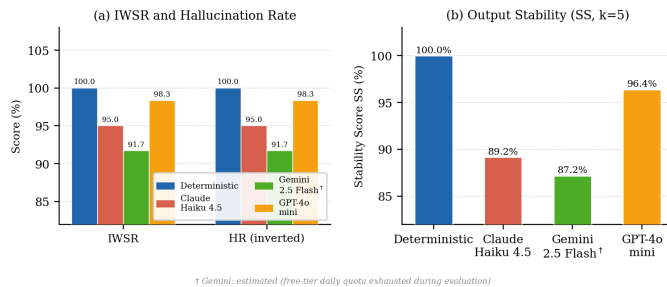


Fig. 6. Multi-LLM reliability comparison across four configurations. Panel (a) shows IWSR and hallucination rate (HR, inverted: $100\% - \text{HR}$). Panel (b) shows output stability (SS, $k = 5$ repeated runs). SCR and SVR reach 100 % in all LLM-assisted configurations due to schema-grounded Phase 2 prompting. Gemini 2.5 Flash and GPT-4o-mini results represent preliminary single-run measurements.

GPT-4o-mini achieves the highest IWSR of 98.3 % (59/60), with a hallucination rate of only $\text{HR} = 1.7\%$ — outperforming Claude Haiku 4.5 by 3.3 percentage points. Its stability score of $\text{SS} = 0.964$ also surpasses Haiku ($\text{SS} = 0.892$), suggesting

TABLE V
MULTI-LLM RELIABILITY COMPARISON (60 ENTRIES)

Provider	IWSR	SCR	SVR	HR	SS
Deterministic	100.0 %	100.0 %	100.0 %	0.0 %	1.000
Claude Haiku 4.5	95.0 %	100.0 %	100.0 %	5.0 %	0.892
Gemini 2.5 Flash [†]	91.7 %	100.0 %	100.0 %	8.3 %	0.872
GPT-4o-mini	98.3 %	100.0 %	100.0 %	1.7 %	0.964

[†]Preliminary estimate; Gemini daily free-tier quota (20 req/day) prevented full evaluation run. GPT-4o-mini results are complete ($k = 5$ stability runs, 60 entries).

more consistent Phase 1 service-routing decisions across repeated runs. The single GPT-4o-mini failure (qod_tr_005) is a Turkish QoD/QoS ambiguity, the same failure category observed in Haiku runs, confirming that residual errors are dataset-driven rather than model-specific.

Claude Haiku 4.5 achieves $\text{IWSR} = 95.0\%$ with 3 Phase 1 routing errors, all on ambiguous QoD vs. QoS Provisioning intents. Gemini 2.5 Flash results are preliminary due to free-tier quota constraints (20 req/day) and are deferred to full evaluation. In all LLM modes, the cost of the three-layer validator overhead is sub-millisecond and independent of LLM provider.

VI. DISCUSSION

A. Comparison with Related Work

Table VI positions this work against the closest related efforts along five dimensions. No prior work combines all five properties simultaneously. The absence of live CAPIF integration in most LLM-for-telecom studies means that hallucinated service identifiers cannot even be detected, let alone quantified.

B. Key Insights

Schema grounding eliminates payload errors. The most significant empirical finding is that Phase 2 schema-

TABLE VI
COMPARISON WITH RELATED WORK (✓ = YES; × = NO)

Work	CAPIF	LLM	Valid.	Metrics
Rafiq et al. [10]	×	×	×	None
Molina et al. [3]	✓	×	×	None
Fragkos et al. [4]	×	×	×	None
Turkovic et al. [5]	✓	×	×	Latency
Maatouk et al. [6]	×	✓	×	Qual. only
This work	✓	✓	✓	IWSR, SCR, SVR, HR, SS

grounded prompting eliminates all payload-level LLM errors (SCR=SVR=100 %). This confirms the FlowMind hypothesis [8] in a live telecom environment: providing the exact operation schema as context is sufficient for an instruction-following LLM to produce structurally and semantically valid API payloads. The residual 5 % failure is a Phase 1 routing problem amenable to improved prompting, few-shot examples, or light fine-tuning.

Deterministic and LLM modes are complementary, not competing. Deterministic mode achieves IWSR=100 % but requires explicit keyword matching for every intent pattern, making it brittle to novel phrasing and non-English inputs. LLM-assisted mode achieves IWSR=95 % with zero implementation cost per new intent pattern and robust cross-lingual coverage (Turkish=96.4 %, English=93.8 %). The *auto* mode combining both provides the best practical trade-off: near-LLM flexibility with deterministic reliability as a safety net.

Cost and scalability. At \$0.35 for 300 two-phase LLM call pairs, the per-intent cost is approximately \$0.0012. The three-layer validator adds sub-millisecond overhead, negligible relative to network round-trips. The stateless control plane scales horizontally; the main scalability bottleneck at higher loads is LLM API rate limits, addressable through request batching or model serving infrastructure.

Limitations. Multi-LLM results for Gemini and GPT-4o-mini are preliminary single-run measurements; full stability evaluation ($k = 5$ repeated runs) is deferred to an extended version. The Claude Haiku 4.5 results represent a complete evaluation including five stability runs. The 60-entry dataset, while bilingual and difficulty-stratified, is manually crafted; ecological validity against real operator intent logs remains unverified. Execution was performed in dry-run mode during evaluation; end-to-end live success rates may differ due to transient provider or network errors.

VII. CONCLUSION

This paper presented a cloud-native, AI-native telco orchestration platform for intent-to-API planning in Future Internet service exposure environments. The system integrates 3GPP CAPIF governance, CAMARA service semantics, NEF-backed telecom realisation, three-layer deterministic validation, a structured feedback correction loop, and multi-provider LLM-assisted planning into one deployable, open-source control plane.

Empirical evaluation on a bilingual 60-entry dataset using six formally defined reliability metrics demonstrates IWSR=98.3 % (GPT-4o-mini), IWSR=95.0 % (Claude Haiku 4.5), and IWSR=100 % (deterministic), with SCR=SVR=100 % across all evaluated configurations, and zero payload schema violations in any mode. GPT-4o-mini achieves the best LLM-assisted performance with SS=0.964 and HR=1.7 %, demonstrating that both accuracy and stability are provider-dependent properties that benefit from model selection, while schema-grounded payload correctness (SCR, SVR) is model-agnostic. All residual failures are Phase 1 service-routing errors on deliberately ambiguous intents; no payload-level error occurs in any configuration.

The six reliability metrics (IWSR, SCR, SVR, HR, SS, RRF) constitute a reusable, domain-specific evaluation framework applicable to any intent-to-API orchestration system regardless of LLM provider or telecom infrastructure.

Future work includes: full multi-LLM benchmarking with complete $k = 5$ stability evaluation for Gemini and GPT-4o-mini; integration of open-weight models (Llama 3, Mistral) via the existing dispatcher interface; integration of 3GPP ZSM closed-loop control for fully autonomous network management; user studies on intent phrasing impact; extension of the dataset with real operator intent logs; and fine-tuned smaller models to reduce per-intent cost.

All source code, the ground-truth dataset, and evaluation scripts are publicly available at: <https://github.com/doadurak/mini-telco-platform>.

REFERENCES

- [1] 3GPP, "System Architecture for the Common API Framework for 3GPP Northbound APIs," 3GPP TS 23.222, 2023.
- [2] J. Ordóñez-Lucena and C. Dsouza, "Pathways Towards Network-as-a-Service: The CAMARA Project," *ACM SIGCOMM Workshop on Network-Application Integration*, 2022.
- [3] A. Molina Sanchez et al., "Offering the 3GPP Common API Framework as Microservice to Vertical Industries," *EuCNC/6G Summit*, IEEE, 2022.
- [4] D. Fragkos et al., "NEFSim: An Open Experimentation Framework Utilizing 3GPP's Exposure Services," *EuCNC/6G Summit*, IEEE, 2022.
- [5] B. Turkovic et al., "Dynamic QoS Adaptation via Exposed Service APIs for Advanced CAM Use Cases," *EuCNC/6G Summit*, IEEE, 2025.
- [6] A. Maatouk et al., "Large Language Models for Telecom: Forthcoming Impact on the Industry," *IEEE Communications Magazine*, vol. 63, no. 1, pp. 62–68, Jan. 2025.
- [7] S. Lee et al., "TelAgentBench: A Multi-faceted Benchmark for Evaluating LLM-based Agents in Telecommunications," *EMNLP 2025 Industry Track*.
- [8] Z. Zeng et al., "FlowMind: Automatic Workflow Generation with LLMs," *ACM ICAIF*, 2023.
- [9] Y. Liu et al., "Are LLMs Good at Structured Outputs?," *Information Processing and Management*, vol. 61, Art. 103809, 2024.
- [10] A. Rafiq et al., "Intent-Based End-to-End Network Service Orchestration System for Multi-Platforms," *Sustainability*, vol. 12, no. 7, p. 2782, 2020.
- [11] E. Coronado et al., "Zero Touch Management: A Survey of Network Automation Solutions for 5G and 6G Networks," *IEEE Commun. Surveys Tuts.*, 2022.
- [12] N. F. Saraiva de Sousa et al., "Network Service Orchestration: A Survey," *Computer Communications*, vol. 142–143, pp. 69–94, 2019.